MIL-E-1/1624A(NAVY)
3 March 1975
SUPERSEDING
MIL-E-1/1624(NAVY)
31 October 1966
(See note 27)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 8682

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Inverted coaxial turnable frequency, 32,900 to 33,500 MHz, 50 kW (minimum) peak power, pulse type

ABSOLUTE RATINGS:

Parameter:	Ef	If(surge)	tk	epy	VSWR	ib	Pi	pi	Du
Unit:	V	A	sec	kv	---	a	W	kw	---
Maximum:	18	15	---	20	1.3:1	20	200	400	0.0005
Minimum:	---	---	300	---	---	12	---	---	---

ABSOLUTE RATINGS:

Parameter:	tpc	rrv	T(anode)	Turner drive torque	Pressurization output
Unit:	µs	kv/µs	°C	in.-oz	psia
Maximum:	2.2	150	150	50	45
Minimum:	0.1	100	-65	---	---
	(Note 6)		(Note 3)	(Note 2)	(Note 4)

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Cooling:	Forced air (note 8)
Mounting position:	Any	Handling:	See note 9
Mounting support:	See figure 1	Output Waveguide:	WC59
Coupling:	See figure 2 (note 10)	Cathode connector:	Jettron 90-006, or equivalent
Weight:	17-1/2 lbs (approx) 7.94 kg		

TEST CONDITIONS:

Parameter:	Ef	epy	VSWR	tk	ib	Du	tpc	rrv	T(anode)
Unit:	V	kV	---	sec	a	---	µs	kv/µs	°C
Maximum:	13.9	19	1.1:1	---	16.7	0.00042	0.22	---	140
Minimum:	12.5	17	---	300	16.2	0.00038	0.18	---	-55
								(Note 6)	

	Frequency
F1	32,900 MHz
F2	33,000 MHz
F3	33,200 MHz
F4	33,400 MHz
F5	33,500 MHz

GENERAL:

Qualification - Required

8682

FSC 5960

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
4027	Temperature coefficient	3	T(anode) = 70° to 140°C; F= F3	ΔF/ΔT	---	0.30	MHz/°C
4030	Thermal frequency drift	23		ΔF	---	30	MHz
1031	Variable-frequency vibration (operating)	12,13		---	---	---	---
4223	Mechanical tuning range	15,16		---	---	---	---
4223	Mechanical tuning fatigue	20,21		---	---	25,000	Cycles
4266	Direct-interelectrode capacitance	-		C	22	28	pF
1047	Low-temperature operation	18,19	T(anode) = 55°C; F = F3	MP	---	0.5	%
	Quality conformance inspection, part 1	11					
4003	Pressurizing	25	40 to 45 psia output assembly	---	---	---	---
4289	Heater current, nonoperating	-	Ef = 16.0 V	If	4.6	5.4	A
4223	Tuner drive torque	20	Ambient temperature	Torque	---	5	in.-oz
4223	Mechanical tuning	15,16, 26	T(anode) = 70° to 110°C	F	32,900	33,500	MHz
4303	Warmup time (heater cathode)	5		tk	300	---	sec
4304	Pulse characteristics	6,7		---	---	---	---
1256	Electrode current (anode)	-	Ib = 6.5 mAdc	---	---	---	---
4250	Power output (1)	-	F = F2, F3, and F4	Po	20	30	W
4250	Power output (2)	-	F = F1 and F5	Po	16	30	W
4308	RF bandwidth	18	F = F1 thru F5	BW	---	2.4/tpc	MHz
4308	Minor lobes	18	F = F1 thru F5	Ratio	7	---	dB
3261	Time jitter	-	F = F1 thru F5	tj(p-p)	---	10	ns
4315	Stability	18	Ib = 6.5 mAdc; F = F1 thru F5	MP	---	0.25	%
4306	Pulse voltage	-	F = F3	epy	17.0	19.0	kv
4280	Electronic tuning range	1	T(anode) = 90°C	ΔF	---	±20	MHz
4223	Resettability	17	33,200 ±50 MHz	ΔF	---	10	MHz
	Quality conformance inspection, part 2						
1143	Forced cooling	3,8,22		T	---	100	°C
1042	Shock, specified pulse	14	11G; t = 11 ms	---	---	---	---
4310	Frequency pulling figure	-	F = F3; VSWR = 1.2:1 (minimum)	ΔF	---	15	MHz
4311	Frequency pushing figure	-	ib = 14 to 16a	ΔF	---	1.0	MHz/a

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3						
---	Life test	24	Group D; VSWR = 1.2 minimum cycled through 26 in 15 minutes	t	1,000	---	hrs
---	Life-test end points:						
4250	Power output (1)	-	F = F2, F3, and F4	Po	16	---	W
4250	Power output (2)	-	F = F1 and F5	Po	14	---	W
4308	RF bandwidth	18	F = F1 thru F5	BW	---	2.5/tpc	MHz
4308	Minor lobes	18	F = F1 thru F5	Ratio	6	---	dB
4315	Stability	18	F = F1 thru F5	NP	---	0.5	%

NOTES:

1. After thermal frequency stability has been reached at each of the frequencies of this test, F1, F2, F3, F4, F5, the curve of frequency versus tuner position shall not deviate more than the amount specified herein from the tuning curve, figure 3, when turning the dial in either direction.

2. The tuning drive to the tube's tuner shaft shall be capable of supplying a minimum of 7 inch-ounces of torque (including inertial effects) when driven to the tube's internal mechanical stops below F1 and above F5.

3. The temperature shall be measured at the point indicated on figure 1, only after thermal equilibrium has been reached.

4. Pressurization of the input bushing is not required at altitudes below 10,000 feet.

5. Prior to the application of anode voltage, the cathode shall be heated to the require initial operating temperature. This may be done by applying 16.0 volts ±5 percent for 300 seconds minimum. Upon the application of anode voltage, the heater voltage shall be lowered to 13.2 volts ±5 percent when the tubes average current reaches 6.5 mAdc.

6. The characteristics of the applied pulse shall be those which result in proper starting and oscillation. The rate of pulse voltage rise, the percentage of pulse voltage ripple, and the rate of pulse voltage fall are among the more important considerations. See method 4304, figure 4304-2.

tpc μs	rrv (note 7) kv/μs	
	Maximum	Minimum
0.1	150	100
1.0	130	100
2.2	120	100

7. No spike or ripple shall exceed ±5 percent of the average peak value of voltage or current. The time of fall of current measured between 20 percent and 85 percent of the smooth peak value of current shall not exceed 0.05 μs. Current pulse droop shall not exceed 5 percent.

8. An airflow of 20 cfm at 760 mmHg shall be directed on the cooling grooves.

9. In handling and mounting the magnetron, care shall be exercised to prevent demagnetization. A minimum distance of 6 inches (152.4 mm) shall be maintained between the tube and ferromagnetic materials or energized magnets.

10. The tube's output is circular waveguide. Transformation from circular to rectangular waveguide shall be accomplished by a mode transition. (See figure 2.)

11. The combined AQL (percent defective) for quality conformance inspection, part 1, shall be 1.0, inspection level II.

12. The tube shall be mounted in a resonance-free jig and vibrated while operating under the conditions specified herein in each of three mutually perpendicular planes (see figure 1) with the vibration frequency varying between 10 and 2,000 Hz at a double amplitude of 0.06 inch (1.52 mm) or 5G whichever is the limiting value. The rate of frequency change shall be logarithmic and shall be such that a complete cycle (10 to 2,000 to 10) will consume approximately 19 minutes. The test shall continue for 120 minutes in each of the three planes. The tube shall be operated continuously throughout the duration of the test.

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NOTES: -Continued

13. After subjection to the specified test, the sample shall meet all requirements of quality conformance inspection, part 1.

14. The sample shall be mounted on a test plate and subjected to 18 impact shocks of the acceleration and duration specified herein. The velocity change shall not be less than 2.5 ft/sec^2. Three shocks shall be applied in each direction along each plane specified on figure 1. No physical change shall result and the sample shall meet the requirements of quality conformance inspection, part 1, mechanical tuning range, and resettability.

15. Nominal tuner dial readings shall be determined corresponding to code letters specified herein (F1 through F5), and the tube shall be marked as specified on figure 1. See figure 3 for tuning curve.

16. With the tuning dial adjusted for a frequency of F3, an operation frequency of F1 will be obtained by counter-clockwise rotation (see figure 1) of the tuning shaft 11 ±1 turns. A frequency of F5 will be obtained by clockwise rotation of the tuning shaft 10 ±1 turns from the F3 setting.

17. The frequency obtained by turning the tuner to a given dial setting in one direction shall be reproducible within the limits specified herein when returning to that same setting from the opposite direction after thermal equilibrium.

18. The rf bandwidth, minor lobes, and stability shall be within the limits specified herein when a VSWR of 1.2:1 minimum is introduced in the load at a distance of approximately 0.2 meter from the tube's output flange, the phase being adjusted for the worst spectrum and instability.

19. Stability shall be measured in terms of the number of output pulses missing, expressed as a percentage of the number of input pulses applied during the period of observation. The missing pulses (MP) due to any causes, are considered to be missing if the rf energy is less than 70 percent of the normal energy level. The VSWR shall be adjusted to that phase producing maximum instability, and the MP counted during the last 3 minutes of a 15-minute test period.

20. The torque required to operate the tuning shaft through the tuning range specified herein shall not exceed the maximum torque specified herein.

21. The tuner shall operate for at least the number of cycles specified herein. The tube shall then meet all the conditions of torque, resettability, and mechanical tuning range at ambient temperature and shall not exceed amount specified herein. One cycle is defined as a complete traverse from one end of the tuning range to the other and back again.

22. With the cooling airflow specified herein, the temperature of the anode (see figure 1) shall not exceed 80°C above the ambient temperature or the maximum specified herein.

23. The tuner dial shall be set for frequency F3. Cooling air shall be applied so that the frequency determining element temperature shall fall between 80°C and 120°C after thermal equilibrium has been reached. The tube shall then be allowed to cool with no voltages applied for at least 15 minutes before the following test. Anode voltage as specified shall be applied after 300 seconds heater warmup time. The frequency as measured at any time within a 20-minute period after application of pulse voltage shall be within the limit specified herein from the frequency after thermal equilibrium has been reached.

24. Life-test shall be conducted in accordance with the following cycle:

Condition	Ib	Ef	Duration (minutes)
Standby	0	16.0	5
Operating	6.5	13.2	60
Off	0	0	5

The accumulation of the "on" time shall be the only time considered in determining compliance with the minimum value. The frequency shall be changed a minimum of 100 MHz every 72 hours during life test between the limits of F2 and F4.

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NOTES: -Continued

25. The pressure specified herein shall be applied to the tube output. There shall be no leaks as evidence by metered pressure fall-off, or by bubbles if the test is performed with the tube immersed in water. The time of the test shall be 1-minute minimum.

26. Normal TUT operation shall be exhibited over the full frequency range from F1 through F5.

27. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-N039)

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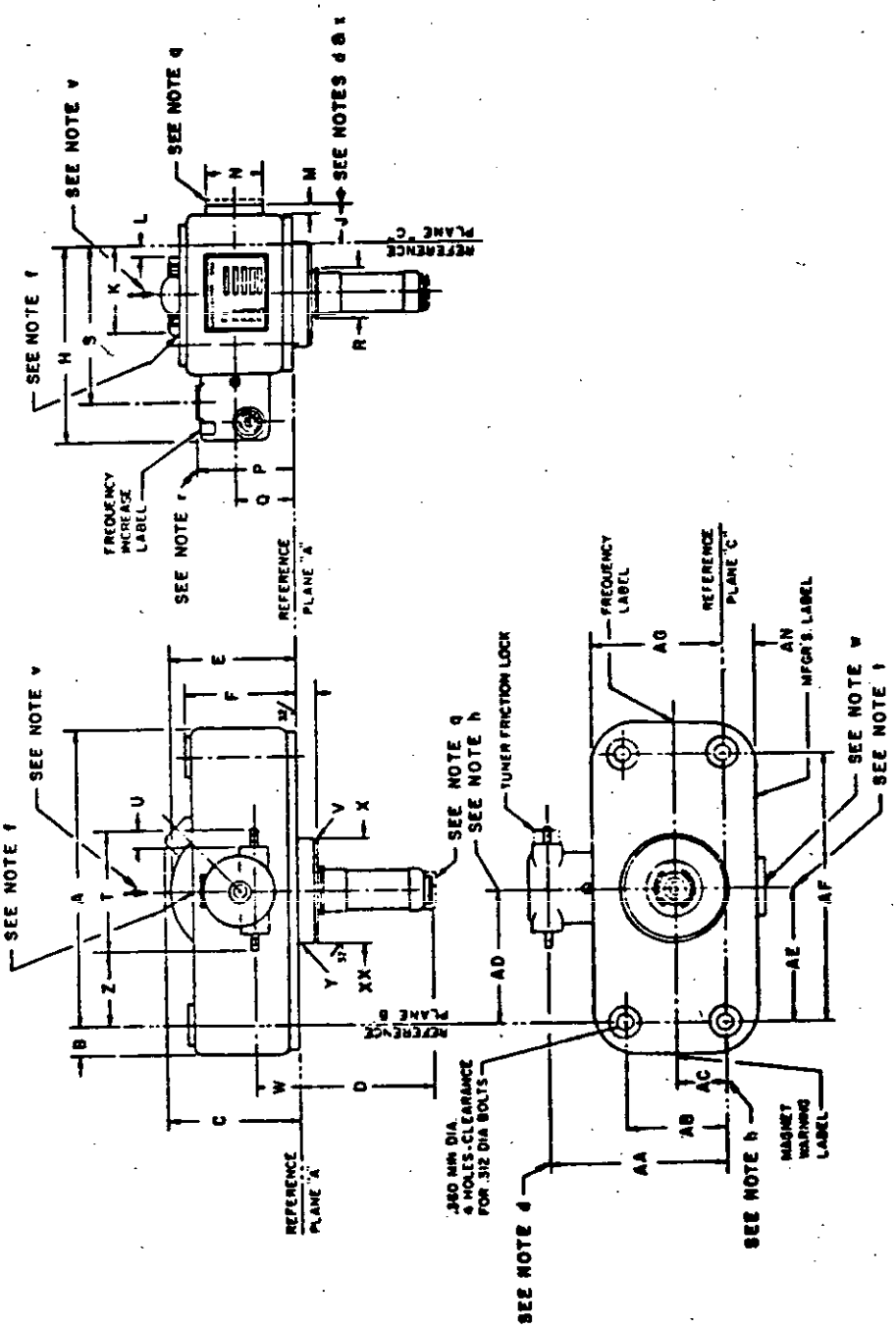


FIGURE 1. Outline drawing of electron tube type 8682.

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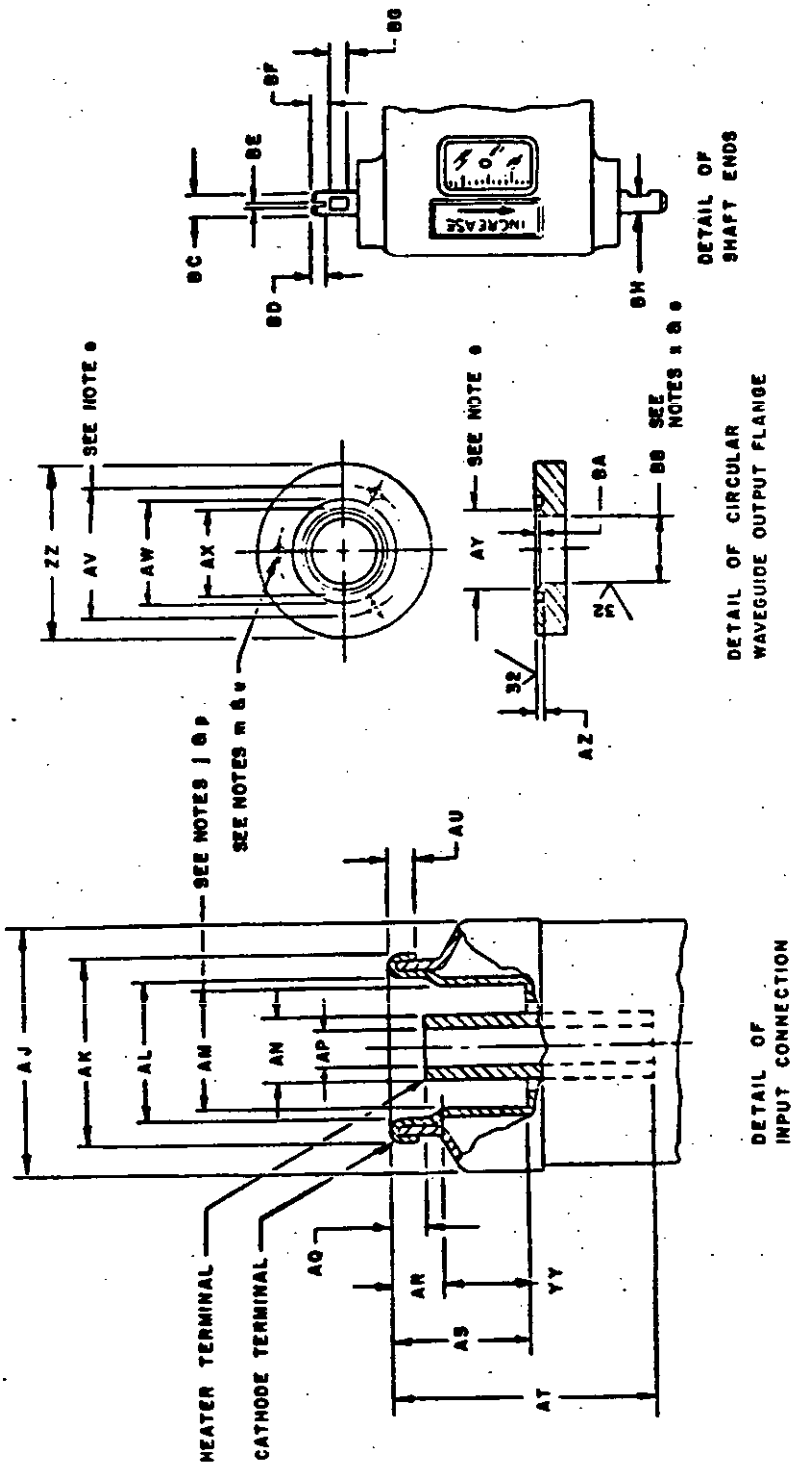


FIGURE 1. Outline drawing of electron tube type 8682 - Continued.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses			
	Minimum		Maximum	
Qualification inspection				
A			7.750	(196.85)
B			.800	(20.32)
C			3.562	(90.47)
D			3.625	(92.08)
E			3.250	(82.55)
H			5.145	(130.68)
AG			3.437	(87.30)
AH			.800	(20.32)
AJ			1.125	(28.58)
AR			.200	(5.08)
AS	.562	(14.27)		
AT	1.000	(25.40)		
R			1.340	(34.04)
Quality conformance inspection, part 1 (see note 11)				
F	2.925	(74.30)	2.945	(74.80)
J	.875	(22.23)	.975	(24.77)
Q	1.490	(37.85)	1.510	(38.35)
U	.428	(10.87)	.448	(11.38)
W	1.070	(27.18)	1.194	(30.33)
X	2.747	(69.77)	2.751	(69.88)
AA	4.566	(115.98)	4.646	(118.01)
AB	2.618	(66.50)	2.632	(66.85)
AC	1.304	(33.12)	1.320	(33.53)
AD	3.492	(88.70)	3.508	(89.10)
AE	3.485	(88.52)	3.515	(89.28)
AF	6.990	(117.55)	7.010	(178.05)
AK	.825	(20.96)	.838	(21.29)
AL	.603	(15.32)	.617	(15.67)
AM	.532	(13.51)	.545	(13.84)
AN	.234	(5.94)	.266	(6.76)
AP	.164	(4.17)	.174	(4.42)
AQ	.125	(3.18)	.187	(4.75)
AU	.115	(2.92)	.135	(3.43)
AV	1.121	(28.47)	1.129	(28.68)
AY	.7005	(17.793)	.7035	(17.869)
BA	.024	(.61)	.028	(.71)
BB	.5925	(15.05)	.5955	(15.126)
BC	.1869	(4.747)	.1873	(4.757)
BD	.120	(3.05)	.130	(3.30)
BE	.038	(.97)	.042	(1.07)

FIGURE 1. Outline drawing of electron tube type 8682 - Continued.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
G	.490 (12.45)	.510 (12.95)
T	3.180 (80.77)	3.194 (81.13)
Y		.060R (1.52)R
AW	.900 (22.86)	.910 (23.11)
AX	.731 (18.57)	.739 (18.77)
AZ	.046 (1.17)	.054 (1.37)
BF	.151 (3.84)	.161 (4.09)
BG	.182 (4.62)	.192 (4.88)
BH	.153 (3.89)	.159 (4.04)
Reference dimensions		
K	2.285 (58.04)	
L	.235 (5.97)	
M	.250 (6.35)	
N	1.500 (38.10)	
P	2.500 (63.50)	
S	4.106 (104.29)	
V	.060 x 45° Chamfer (1.52 x 45°)	
Z	1.907 (48.44)	

NOTES:

a. Reference plane "A" is defined as a plane passing along the face of the mounting plate.
b. Reference plane "A" is defined as a plane perpendicular to plane "A" passing through the axis of the holes, as shown at reference plane "A".
c. Reference plane "C" is defined as a plane mutually perpendicular to planes "A" and "B" passing through the axis of the holes, as shown at reference plane "A".
d. Includes angular as well as lateral deviations.
e. These diameters shall be concentric within 0.002 T.I.R.
f. Anode temperature shall be measured at this point.
g. For vibration and shock testing, the planes of testing shall be reference planes "A", "B", and "C".
h. This dimension applies to diameter "XX".
j. This diameter shall be concentric with diameter "XX" within 0.030 inch.
k. Heater terminal and cathode terminal shall be concentric within 0.010 inch.
l. Input connection mates with Jettron Products, Inc. connectors, No. 90-006 and 90-030, or equivalent.
m. Four (112) - 40 NC 3 holes spaced 120 degrees, 30'/119 degrees, 30' apart, by 0.250 inch deep.
n. Mates with circular waveguide, EIA designation WC-59, or equivalent.
p. This diameter applies for length of "YY".
q. Protective closures.
r. For pressurization of the input, use an AMS 3303-148 "O" ring, or equivalent, sealed to diameter "XX".
s. For pressurization of the output, use an AMS 3303-18 "O" ring, or equivalent.
t. Applies to diameter "ZZ" of output flange.
u. A plane passing through the axis of this hole and the axis of diameter "ZZ" shall be parallel with reference plane "B" within 0.010 inch. This hole shall be located 180 degrees from the input connection.
v. Indicates direction of anode cooling airflow.
w. South pole.
x. This surface of the output flange shall be parallel with reference plane "C" within 0.005 T.I.R.

FIGURE 1. Outline drawing of electron tube type 8682 - Continued.

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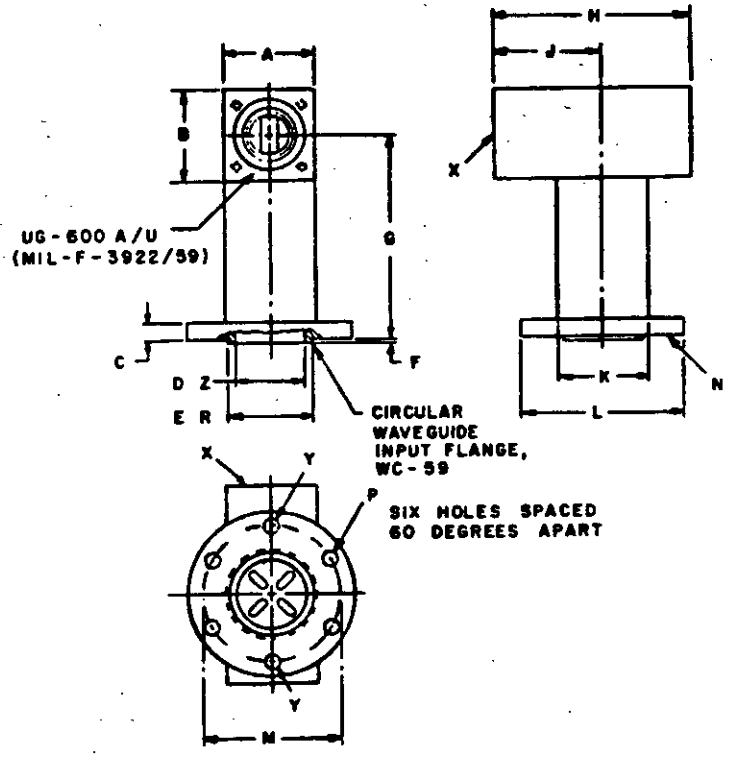


FIGURE 2. Circular to rectangular transition.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
A	.736 (18.69)	.764 (19.41)
B	.736 (18.69)	.764 (19.41)
C	.143 (3.63)	.157 (3.99)
H	1.673 (42.49)	1.687 (42.85)
K	.738 (18.75)	.762 (19.35)
L	1.365 (34.67)	1.385 (35.18)
M	1.122 (28.50)	1.128 (28.65)
Quality conformance inspection, part 1		
E	.698 (17.73)	.699 (17.75)
F	.020 (.51)	.024 (.61)
G	1.679 (42.65)	1.693 (43.00)
J	.891 (22.63)	.909 (23.09)
P	.120 (3.05)	.130 (3.30)
Quality conformance inspection, part 2		
D	.590 (14.99)	.602 (15.29)

NOTES:

a. Diameter "R" shall be concentric with reference diameter "Z" within 0.004 T.I.R.
b. Surface "N" shall be perpendicular to surface "X" within 0.010 T.I.R.
c. A plane passing through the axis of diameters "Y" at surface "N" shall be perpendicular to surface "X" within 0.010 T.I.R.
d. For pressurization of the choke flange, use an AMS 3303-13 "O" ring, or equivalent.
e. For pressurization of the circular waveguide flange, use an AMS 3303-18 "O" ring, or equivalent.
f. All surfaces other than the flange faces shall be painted semi-gloss gray conforming to FED-STD-595, color 26373.
g. Electrical parameters:
   (1) Center frequency - 33.2 ±0.3 GHz.
   (2) VSWR - 1.15 (maximum) over frequency range of 32.9 to 33.5 GHz.
   (3) Mode purity - 20 dB (minimum) over frequency range of 32.9 to 33.5 GHz.
   (4) Pressurization - 40 psig (maximum).
   (5) Power handling capability = 100 kW peak 40 watt average at 15 psig dry air.
   (6) Insertion loss - 0.3 dB (maximum).
   (7) Pressure leakage - There shall be no bubbles when immersed in water and pressurized up to 40 psig for 1 minute.

FIGURE 2. Circular to rectangular transition - Continued.

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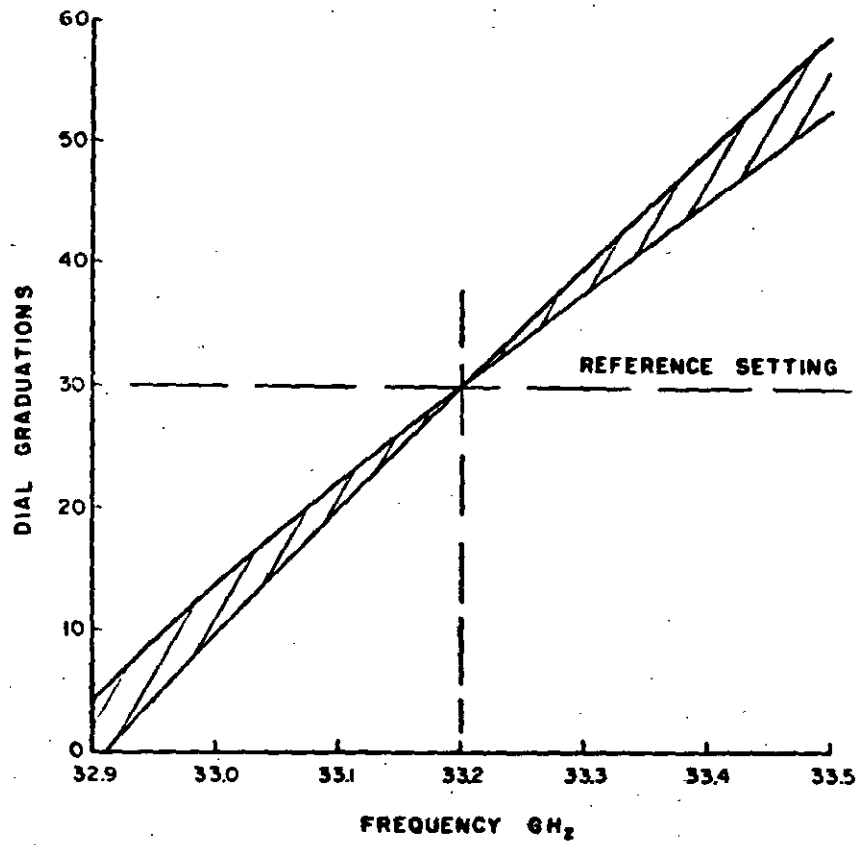


FIGURE 3. Tuning curve.

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